A Comprehensive Reliability Assessment of hBN-encapsulated MoS_2 FETs and Inverters

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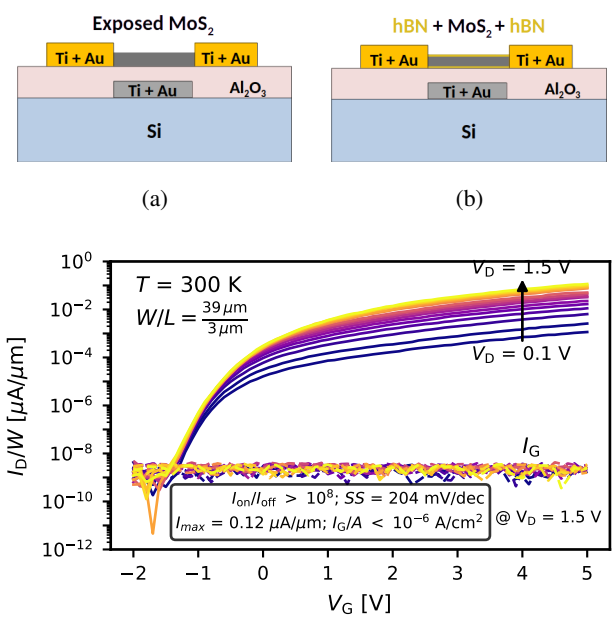
I. Introduction

II. Methods and Fabrication



(a) (b)

(c)

Fig. 1. **Cross-section schematics**: (a) exposed and (b) hBN-encapsulated MoS_2 FETs. (c) **Performance of the fabricated devices**: $I_{D(G)}$-V_G curves for different drain voltages and main device metrics.

Acknowledgment

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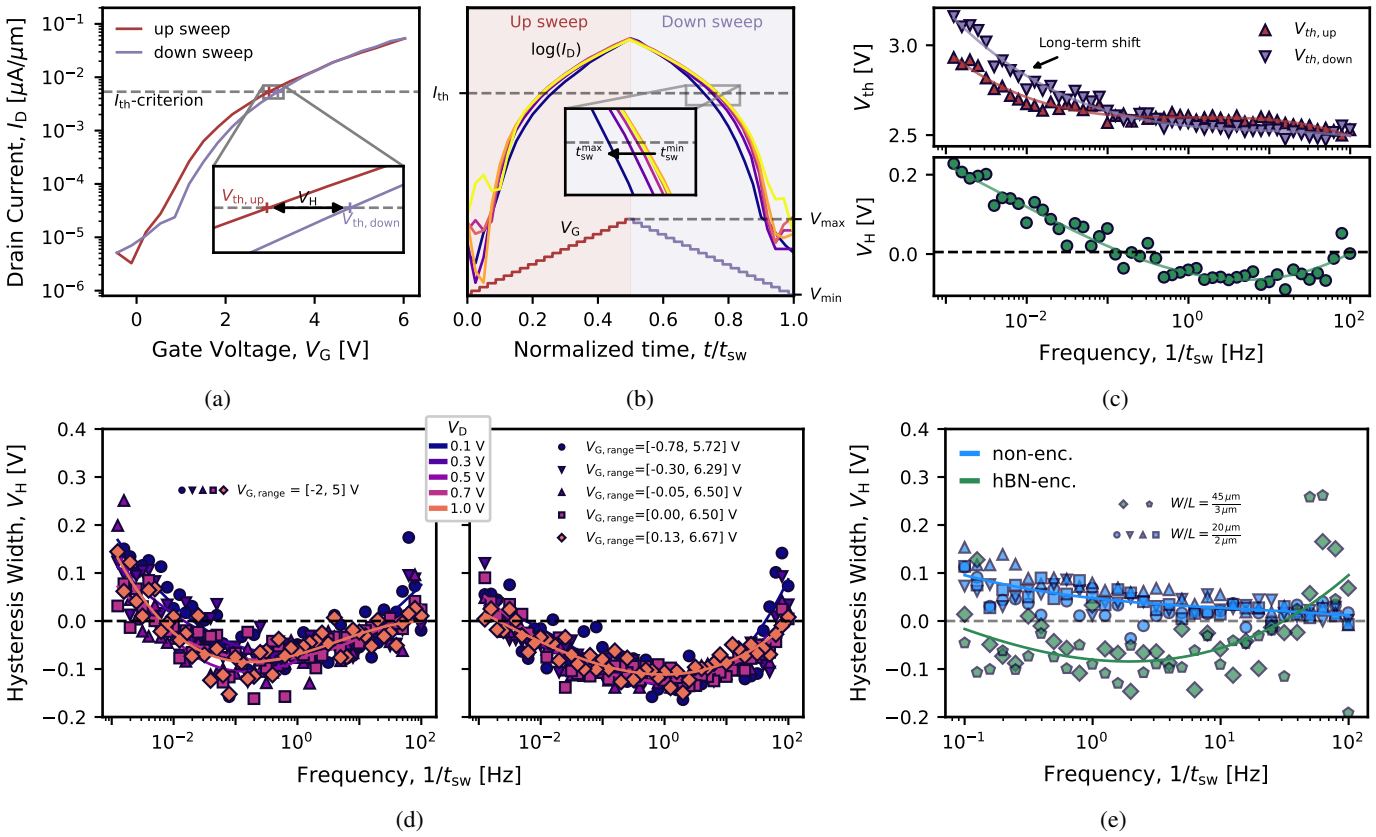

Fig. 2. (a) **Typical hysteresis curve**: the hysteresis width, V_{H}, is defined as the difference between the threshold voltages extracted during the up and down sweeps, i.e., $V_H = V_{\mathrm{th,up}} - V_{\mathrm{th,down}}$. (b) **Hysteresis dependence on the gate voltage sweep rate**: $\log(I_D)$ and V_{G} are shown as a function of the normalized sweep time, t/t_{sw}, for different sweep rates, r_{sw}. (c) **Hysteresis-r_{sw} curves**: Threshold voltages, $V_{\mathrm{th,up}}$ and $V_{\mathrm{th,down}}$, and hysteresis width, V_{H}, are shown as a function of the sweep rate, r_{sw}.

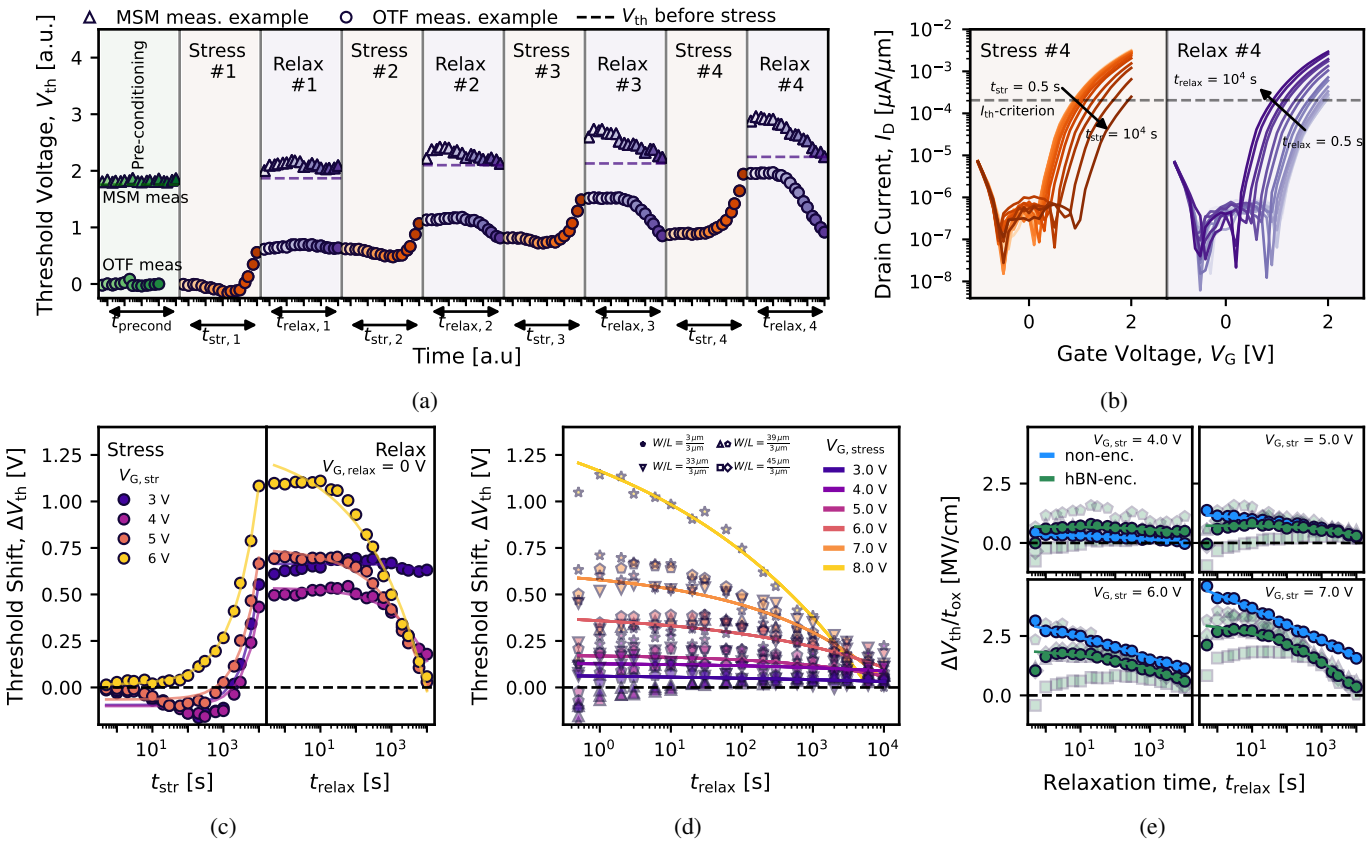

Fig. 3. (a) Example of a hysteresis curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.